

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY



Lecture No. 7



By- SATISH YADAV SIR

TOPICS TO BE
COVERED

01 Basics of relations

02 Types of relations

03 Number Of relations

Relations

Total no. of Relations = Total no. of subsets.

$$|A| = m \quad |B| = n \quad \underline{2^{m \cdot n}}$$

$$|A| = n \quad \underline{2^{n^2}}$$

$$\# R \rightarrow 2^{n^2 - n}$$

$$\underline{\# \text{sym}} \rightarrow 2^n \cdot 2^{\frac{n^2 - n}{2}}$$

→ no problem with same element
→ demands the flipping.

Relations



Irreflexive: $A \rightarrow$ non empty set

$$\forall a \in A (a, a) \notin R$$

→ hates same elements

$$A = \{1, 2, 3\}$$

$$A \times A \rightarrow \boxed{} \rightarrow R_1$$

$$R_1 = \{(\underline{1, 1}), (1, 2)\}$$

$$\forall a \in A (a, a) \notin R$$

→ not irreflexive

Relations



$$R_1 = \{ (12) (21) (\underline{22}) \}$$

not
Irreflexive(x)

not Reflexive

= Irreflexive
wrong

$$R_2 = \{ (13) (23) (33) \}$$

not Irreflexive
not Reflexive

$$R_3 = \{ \boxed{(11)(22)(33)} (12) \}$$

Reflexive
not Irreflexive

Relations



Reflexive:

needs diagonal elements.

$$R_1 = \{ \boxed{11 \ 22 \ 33} (13) \}$$

$$R_2 = \{ \boxed{11 \ 22 \ 33} (12) \}$$

$$R_3 = \{ | + \}$$

Irreflexive:

hates same elements.

$$R_1 = \{ (11) \mid \times \}$$

$$R_2 = \{ \mid \checkmark \}$$

Relations



Irreflexive:

$$R_1 = \{ (1, 2) \} \checkmark$$

$$R_2 = \{ (1, 2), (2, 3) \} \checkmark$$

$$R_6 = \{ (1, 1), (2, 2), (3, 3), (1, 2) \} \times$$

$$\downarrow$$
$$\nexists a \in A (a, a) \notin R.$$

$$R_3 = \{ \} \checkmark$$

$$R_4 = \{ (1, 2), (2, 3), \boxed{(3, 3)} \} \times$$

$$R_5 = \{ (\overset{\checkmark}{2}, 1), (\overset{\checkmark}{2}, 3), (\overset{\checkmark}{3}, 2), \boxed{(\overset{\times}{2}, 2)} \} \times$$

Relations



$$|A| = n$$

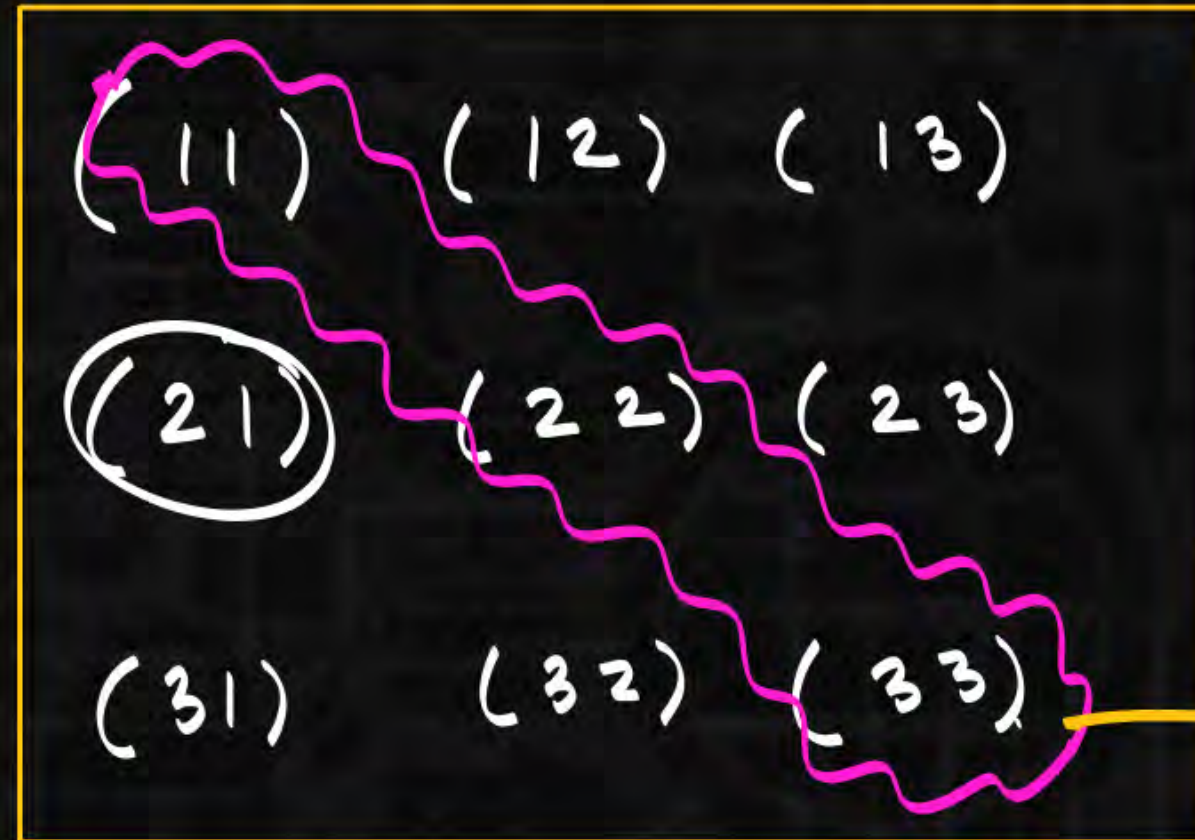
$$A = \{1, 2, 3\}$$

$A \times A =$

Diagonal elements = n .

Total elements = n^2

non diagonals = $n^2 - n$.



non diagonal
↓
2 choices

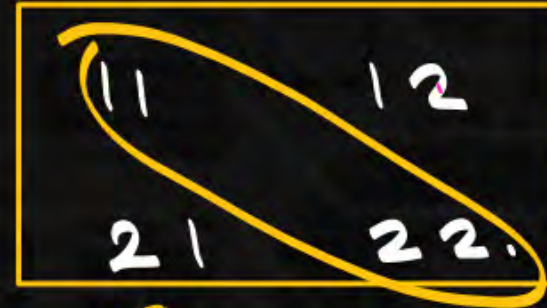
→ 1 choice
(absent)

$$\text{Total no. of Irreflexive} = 2^{n^2 - n}$$

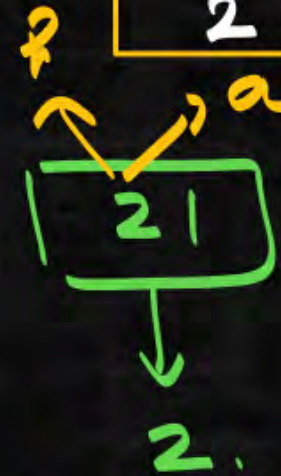
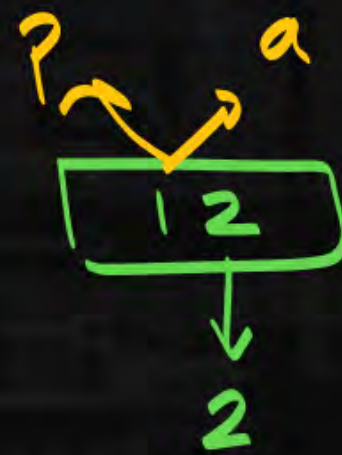
Relations

$$A = \{1, 2\}$$

$$A \times A =$$



choice(absent)



$$\begin{matrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{matrix}$$

$$\longrightarrow R_1 = \{ \} \checkmark$$

$$R_2 = \{ (2, 1) \} \checkmark$$

$$R_3 = \{ (1, 2) \} \checkmark$$

$$R_4 = \{ (1, 2), (2, 1) \} \checkmark$$

Relations



Antisymmetric:

$$\forall a \forall b \left((a, b) \in R \wedge (b, a) \in R \longrightarrow a = b \right)$$

$$R_1 = \{ (1, 2) \} \vee (a, b) \in R \wedge (b, a) \in R \longrightarrow a = b$$

$$\boxed{(1, 2) \in R} \wedge \boxed{(2, 1) \in R} \longrightarrow$$

$$\text{F} \longrightarrow \text{T}$$

Relations

$$R_1 = \{ (1, 2) \} \text{ Anti} \checkmark$$

$$(a, b) \in R \wedge (b, a) \in R \rightarrow a = b$$

$$\begin{array}{c} \underline{(1, 2) \in R} \wedge \underline{(2, 1) \in R} \rightarrow a = b \\ \quad \text{T} \quad \wedge \quad \text{F} \\ \hline \text{F} \rightarrow a = b \end{array}$$

True

Relations



$$R_2 = \{ (1,1) \} \text{ Anti} \checkmark$$

$$(a,b) \in R \wedge (b,a) \in R \longrightarrow a = b.$$

$$(1,1) \in R \wedge (1,1) \in R \longrightarrow 1 = 1.$$

$$a = 1$$

$$b = 1$$

Anti.
no problem
with same
element

Relations

$$R_3 = \{ (1, 2) (2, 1) \} \text{ not Anti}$$

$$(a, b) \in R \wedge (b, a) \in R \longrightarrow a = b.$$

$$\begin{matrix} a = 1 \\ b = 2 \end{matrix}$$

$$\underbrace{\underbrace{(1, 2) \in R}_T \wedge \underbrace{(2, 1) \in R}_T}_{T} \longrightarrow \begin{matrix} 1 = 2. \\ \downarrow \\ \text{f} \end{matrix}$$

false

Relations



Anti $\rightarrow$ hates flipping.
 $\rightarrow$ no problems
with same element.

$$R_1 = \{ (\underline{23}) (\underline{32}) (21) \} \times$$

$$R_2 = \{ (\underline{22}) (\underline{12}) \} \checkmark$$

$$R_3 = \{ (\overset{\checkmark}{\underline{12}}) (\overset{\checkmark}{\underline{11}}) (\overset{\checkmark}{\underline{22}}) \}$$

Relations



$$R_1 = \{ (\underline{11}) (12) \} \checkmark$$

$$R_2 = \{ (12) \} \checkmark$$

$$R_3 = \{ (23) \} \checkmark$$

$$R_4 = \{ (32) \} \checkmark$$

$$R_5 = \{ \cdot \} \checkmark$$

Relations

$$A = \{1, 2, 3\}$$

$$A \times A = \{ \textcircled{(1,1)} \textcircled{(2,2)} \textcircled{(3,3)} \boxed{(1,2)} \textcircled{(2,1)} \boxed{(2,3)} \boxed{(3,2)} \boxed{(1,3)} \boxed{(3,1)} \}$$

Diagram illustrating the Cartesian product $A \times A$ for $A = \{1, 2, 3\}$. The elements are grouped into three categories based on their status:

- Present (Pink circles):** $(1,1)$, $(2,2)$, $(3,3)$, $(2,1)$. Each of these has a pink arrow pointing to the word "present".
- Absent (Yellow boxes):** $(1,2)$, $(2,3)$, $(3,2)$, $(1,3)$, $(3,1)$. Each of these has a yellow arrow pointing to the word "absent".

present

absent

present

- $\{(2,1)\} \checkmark$
- $\{(1,2)\} \checkmark$
- $\{\} \checkmark$

Relations



→ 2 choices.

Diagonal elements.

$$= 2^n.$$

$$\text{non diagonal} = n^2 - n.$$

$$\text{boxes} = \frac{n^2 - n}{2}.$$

$$\text{Total Anti} = 2^n \cdot 3^{\frac{n^2 - n}{2}}.$$

Relations



$$R_1 = \{ (a, b) \mid a+b \leq 3 \}$$

not Anti

$$a R_1 b \wedge b R_1 a \rightarrow a = b$$

$$a+b \leq 3 \wedge b+a \leq 3 \rightarrow a=b$$

$$1+2 \leq 3 \wedge 2+1 \leq 3$$

$$\frac{(1,2) \in R}{T} \wedge \frac{(2,1) \in R}{T} \rightarrow 1=2$$

$\rightarrow F$

$$R_2 = \{ (a, b) \mid a = b+1 \}$$

$$a R_2 b \wedge b R_2 a \rightarrow a = b$$

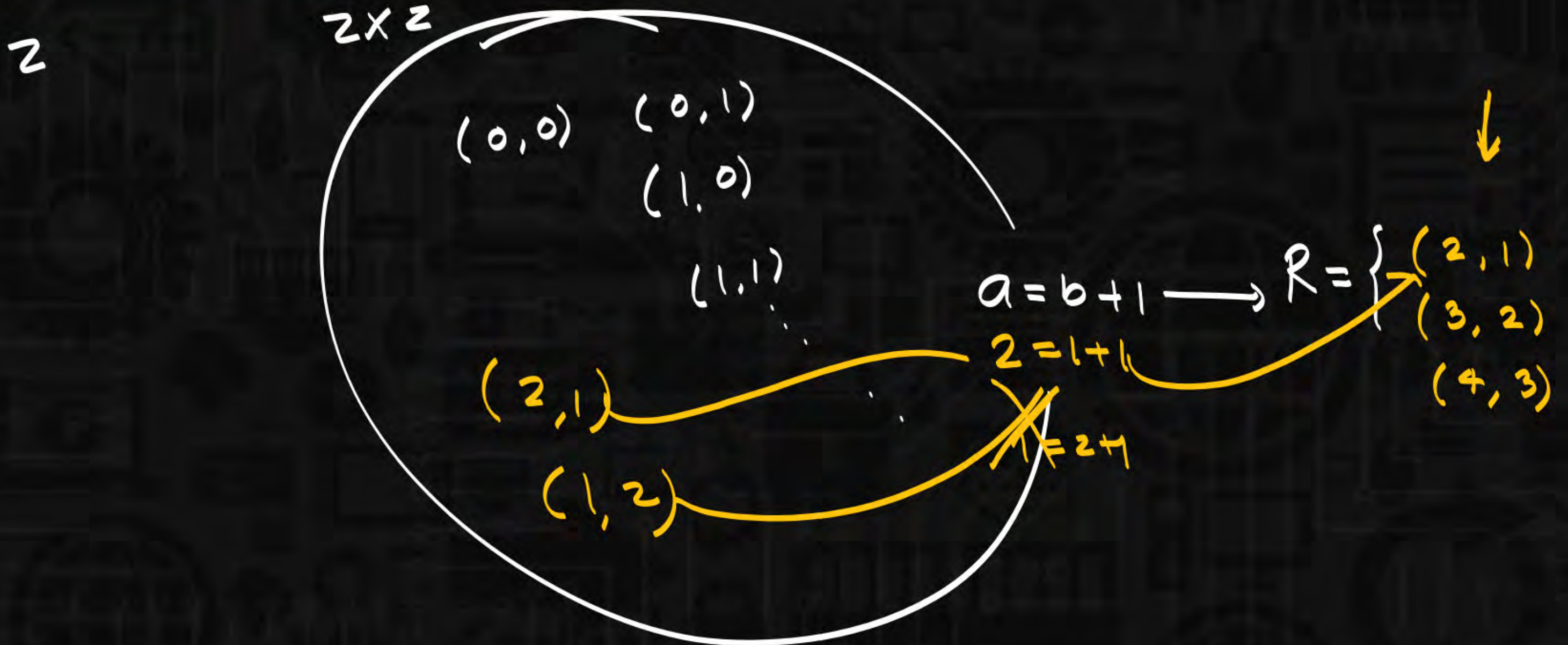
$$\frac{a = b+1}{T} \wedge \frac{b = a+1}{F} \rightarrow a = b$$

$$\text{Case 1: } T \wedge F \rightarrow$$

$$\text{Case 2: } F \wedge T \rightarrow$$

b.

Relations



Relations



Asymmetric:

$$\forall a \forall b ((a, b) \in R \longrightarrow (b, a) \notin R)$$

$$R_1 = \{ \} \checkmark$$

$$(a, b) \in R \longrightarrow (b, a) \in R.$$

$$\frac{\frac{f}{\longrightarrow}}{T}$$

$$R_2 = \{ (1, 2) \} \checkmark$$

$$(a, b) \in R \longrightarrow (b, a) \notin R.$$

$$\frac{(1, 2) \in R}{T} \longrightarrow \frac{(2, 1) \notin R}{T}$$

Relations

$$R_3 = \{ (2, 3) (3, 2) \} \text{ not Asymmetric.}$$

$$(a, b) \in R \rightarrow (b, a) \notin R.$$

$$(2, 3) \in R \rightarrow (3, 2) \notin R.$$

$$\begin{array}{ccc} \text{T} & \rightarrow & \text{F} \\ \hline & & \text{false.} \end{array}$$

Relations



$$R_3 = \{ (1,1) \} \quad \text{not Asymmetric}$$

Asymmetric

$$(a,b) \in R \longrightarrow (b,a) \notin R$$

$\longrightarrow$ no flipping.

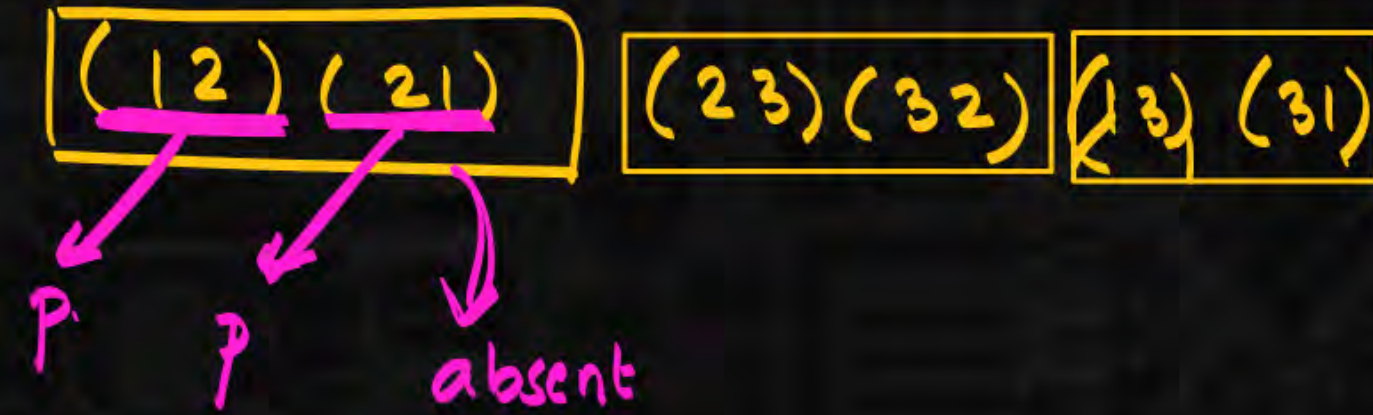
$$\underbrace{(1,1) \in R}_T \longrightarrow \underbrace{(1,1) \notin R}_F$$

$\longrightarrow$ no same element.

$$a=1$$

$$b=1$$

Relations



$$bones = \frac{n^2 - n}{2}$$

$$Asymmetric = \frac{n^2 - n}{2}$$

3

Relations



Symmetric

$$(a, b) \in R \rightarrow (b, a) \in R$$

→ Demands flipping.

→ allows same element.

$$2^n \cdot 2^{\frac{n^2-n}{2}}$$

Antisymmetric

$$(a, b) \in R \wedge (b, a) \in R \rightarrow a = b$$

→ no flipping.

→ allows same element.

$$2^n \cdot 3^{\frac{n^2-n}{2}}$$

Asymmetric

$$(a, b) \in R \rightarrow (b, a) \notin R.$$

→ no flipping.

→ no same element.

$$3^{\frac{n^2-n}{2}}$$

Relations



$$R = \{ \underbrace{(xy)}_{\text{no sy}}, \underbrace{(xz)(zx)}_{\text{not Ant.}}, (zy) \} \quad (\text{GATE-09})$$

not Symmetric $\rightarrow$ no flipping
of (xy)

not Anti $\rightarrow (xz)(zx)$

Relations

$$R = \{ \overset{\text{Anti X.}}{\textcircled{(11)}} (21) (22) (\underline{23}) (\overset{\downarrow}{24}) (31) (\underline{32}) (33) (34) \}$$

Anti X.

$\textcircled{6-98}$

$$A = \{1, 2, 3, 4\}$$

1. Ref $\times \underline{(9,7)} \in R$
2. Sym $(1,2) \notin R$
3. Anti
4. Asym.

Relations

Set: A

C.P $A \times A \longrightarrow \boxed{} \longrightarrow \text{Result}$

3R5
5R7

Set: A^2

C.P $A^2 \times A^2 \longrightarrow \boxed{} \longrightarrow \text{Result}$
 $(3,5) R (6,5)$

Relations

$$\mathbb{Z}^2 \times \mathbb{Z}^2$$

$$a R b \rightarrow b R a$$

$$L S R R S \rightarrow R S R L S$$

$$R: \{ (a,b) R (c,d) \mid ad=bc \mid \text{Symm} \}$$

$$\begin{array}{c} \text{LS} \quad R \quad \text{RS} \\ \hline (a,b) R (c,d) \end{array} \xrightarrow{\quad} \begin{array}{c} \text{RS} \quad \text{LS} \\ \hline (c,d) R (a,b) \end{array}$$

$$ad=bc$$

$$\rightarrow cb=da$$

Relations

hw. $A \rightarrow \text{set}$
 $A \times A \rightarrow \mathcal{P}$

R_1, R_2 are reflexive relation of A . $R_1 \cup R_2 = \text{reflexive?}$
 $R_1 \cap R_2 = \text{reflexive.}$

R_1, R_2 } Sym ✓
 R_1, R_2 } Anti
 R_1, R_2 } Asy.
 R_1, R_2 } Ir.

